

● = refereed venue (CoRL, IEEE TETCI); others are preprints

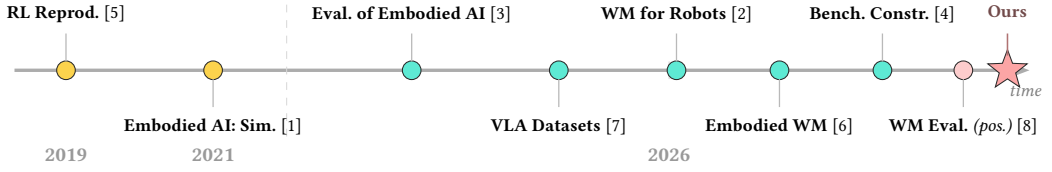


Fig. 1. The closest benchmark/evaluation surveys as a timeline. Two predate the recent wave, an IEEE TETCI survey of embodied-AI simulators and tasks [1] and a CoRL survey of real-robot RL reproducibility [5] (amber = refereed venue); the rest are 2026 preprints on embodied evaluation, VLA data, and world models. None organises the robot-evaluation landscape by evaluation-mode $\times$ capability $\times$ model-family; the sole work reaching the advantage-of-prediction axis is a *position* paper [8]. **Ours** (★) fills that cell; per-axis coverage is in Table ??.

REFERENCES

- [1] Jiafei Duan, Samson Yu, Hui Li Tan, Hongyuan Zhu, and Cheston Tan. 2022. A Survey of Embodied AI: From Simulators to Research Tasks. *IEEE Transactions on Emerging Topics in Computational Intelligence* (2022). arXiv:2103.04918.
- [2] Bohan Hou, Gen Li, Jindou Jia, Tuo An, Xinying Guo, Sicong Leng, Haoran Geng, Yanjie Ze, Tatsuya Harada, Philip Torr, Oier Mees, Marc Pollefeys, Zhuang Liu, Jiajun Wu, Pieter Abbeel, Jitendra Malik, Yilun Du, and Jianfei Yang. 2026. World Model for Robot Learning: A Comprehensive Survey. arXiv:2605.00080.
- [3] Liyu Hou, Linyuan Gao, Yuan Wu, and Yi Chang. 2026. A Survey on Evaluation of Embodied AI. Authorea preprint, DOI:10.22541/au.176851544.45077723.
- [4] Jinshan Lai, Jianwei Hu, Baoyang Jiang, Fengchun Zhang, Leyuan Wang, Haotian Li, Yida Wang, Tingxuan Huang, Xi Ren, and Qiang Ma. 2026. Intelligent Automation for Embodied Benchmark Construction: Pipelines, Embodiments, Simulators, and Trends. arXiv:2606.12207.
- [5] Nicolai A. Lynnerup, Laura Nolling, Rasmus Hasle, and John Hallam. 2020. A Survey on Reproducibility by Evaluating Deep Reinforcement Learning Algorithms on Real-World Robots. In *Conference on Robot Learning (CoRL)*, PMLR vol. 100. arXiv:1909.03772.
- [6] Yu Shang, Yinzhou Tang, Xin Zhang, Shengyuan Wang, Yuwei Yan, Honglin Zhang, Zhiheng Zheng, Jie Zhao, Jie Feng, Chen Gao, Jiansheng Chen, and Yong Li. 2026. A Survey of Embodied World Models. Preprint (non-arXiv).
- [7] Ziyao Wang, Bingying Wang, Hanrong Zhang, Tingting Du, Tianyang Chen, Guoheng Sun, Yexiao He, Zheyu Shen, Wanghao Ye, and Ang Li. 2026. Vision-Language-Action in Robotics: A Survey of Datasets, Benchmarks, and Data Engines. arXiv:2604.23001.
- [8] Yang Yu, Shiyuan Zhang, Yifei Sheng, Haoxiang Ren, and Haoxin Lin. 2026. How Should World Models Be Evaluated for Embodied Decision-Making? A Decision-Making-Centric Position. arXiv:2606.15032.